Applied Bayesian Statistics in Animal & Biological Sciences

5) Discrete data modeling

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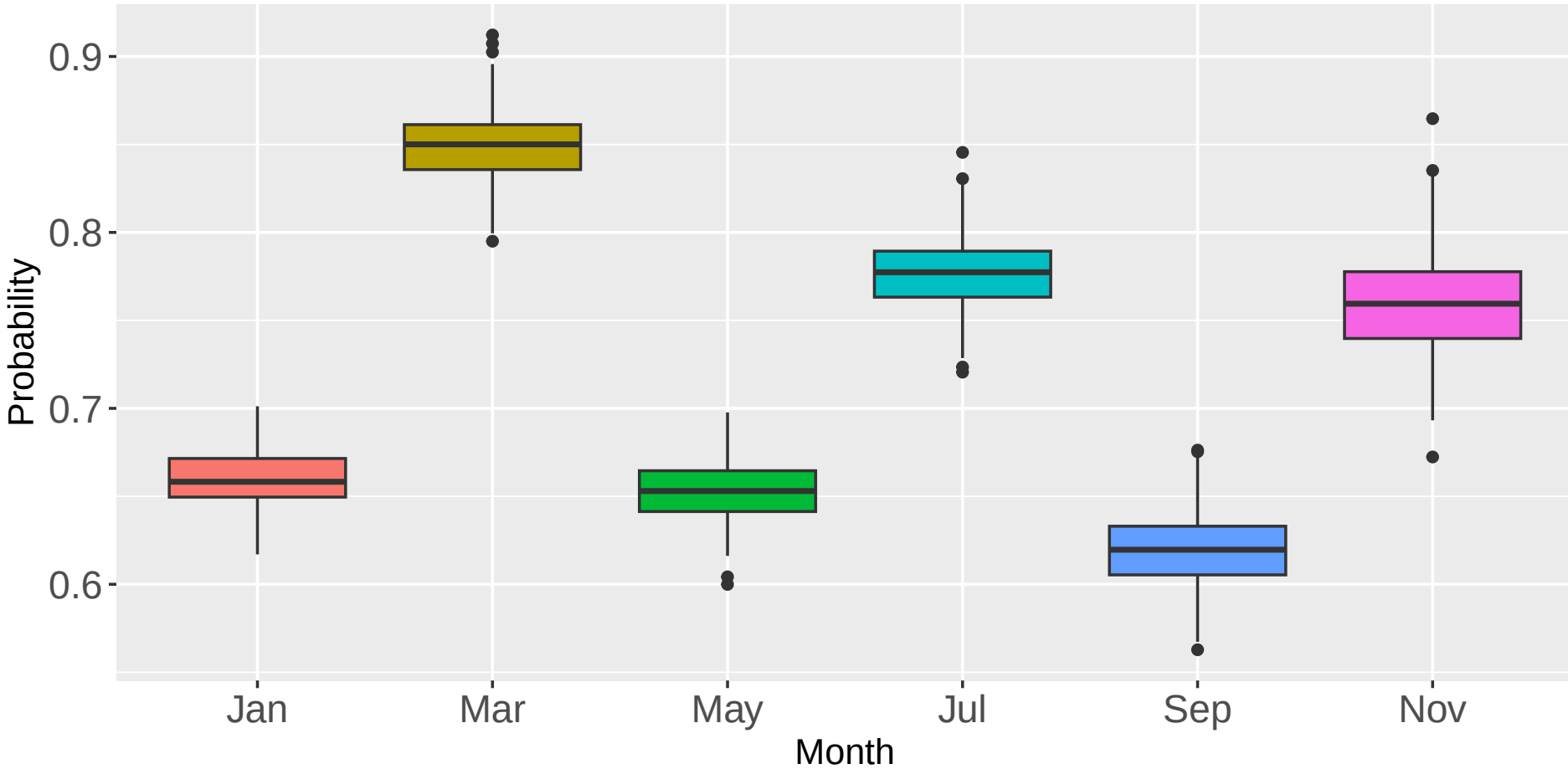
Today's program

- Dataset for chicken vaccination rates against Newcastle disease in rural Tanzania (binomial model + logit-function)
- Traffic incident count data (poisson models, linear prediction, log-link function, over-dispersion)
- Rural Tanzania chicken numbers per household after vaccination dataset (poisson models, over-dispersion and hierarchical modeling)
- The negative binomial distribution for over-dispersed data

Chicken vaccination rates in Tanzania by campaign month



- Binomial modeling: Does month of vaccination campaign affect the relative participation rate?

Binary data modeling of vaccination rates in rural Tanzania

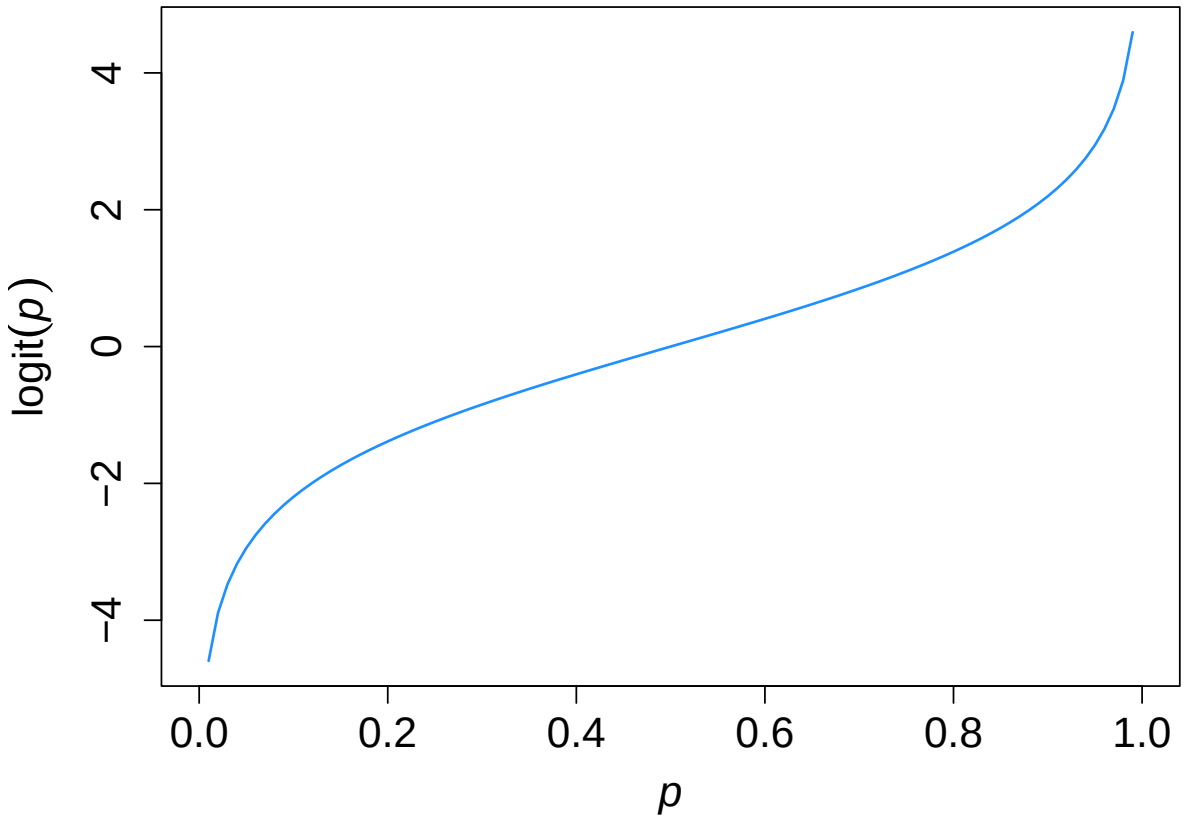
	Jan	March	May	July	Sept	Nov
n	709	301	804	400	674	198
y	468	256	525	310	418	151
Odds ratio	0.66	0.85	0.65	0.78	0.62	0.76

$$y_i \sim \text{Binomial}(n, p) \tag{1}$$

- n trials, and p probability of success (the odd ratios)
- Binary data: 468 times a 1 and 709-468=241 times a 0 in January

De Bruyn et al., (2017)

Linear prediction and binary data (binomial) modeling



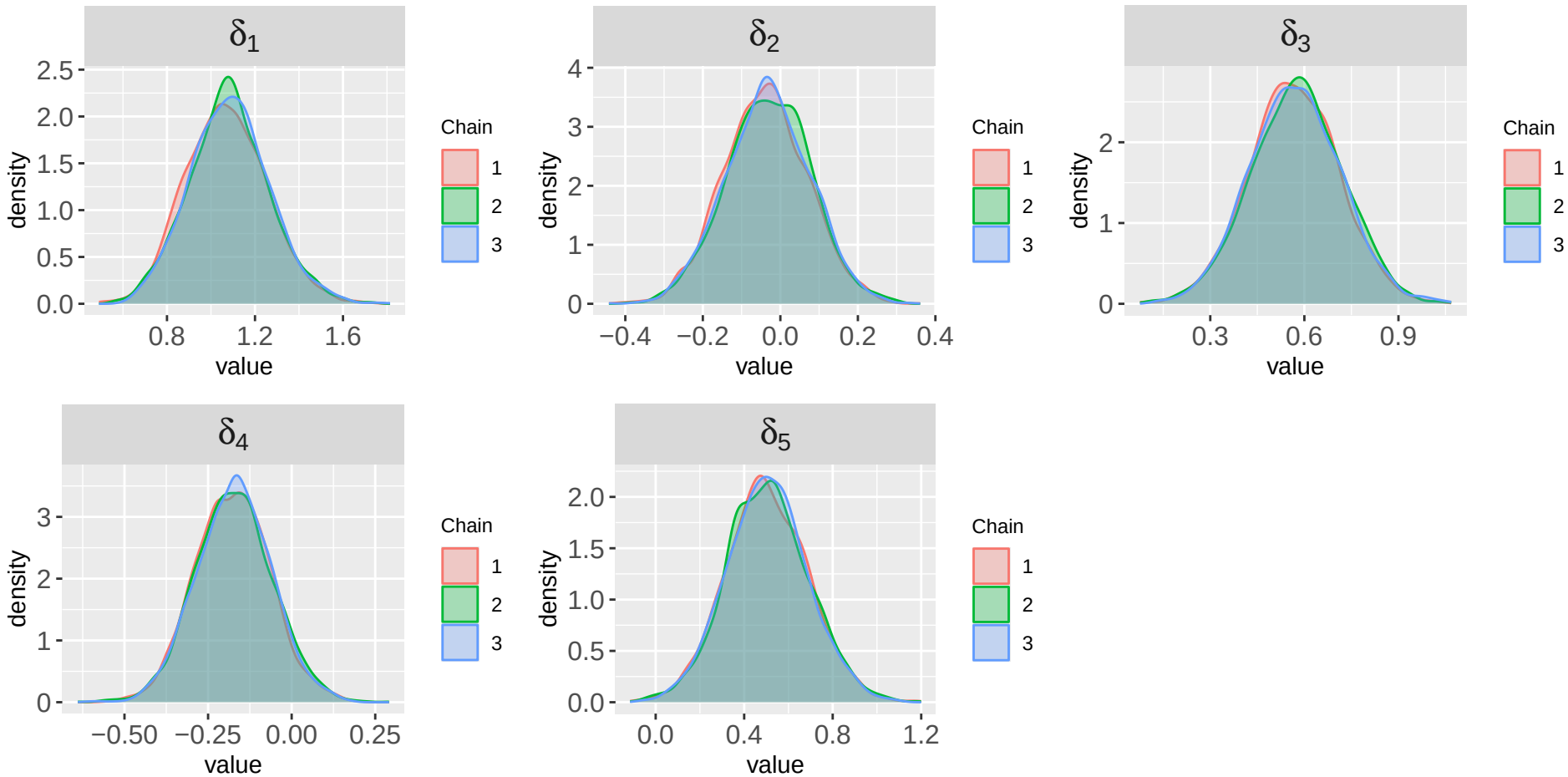
$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \eta_i = \alpha + \beta_{month,i} \tag{2}$$

- The logit function connects (binomial) probabilities ($0 \leq P \leq 1$) to linear predictors $[-\infty, \infty]$.

Logit binomial prediction of vaccination rates in RSTAN

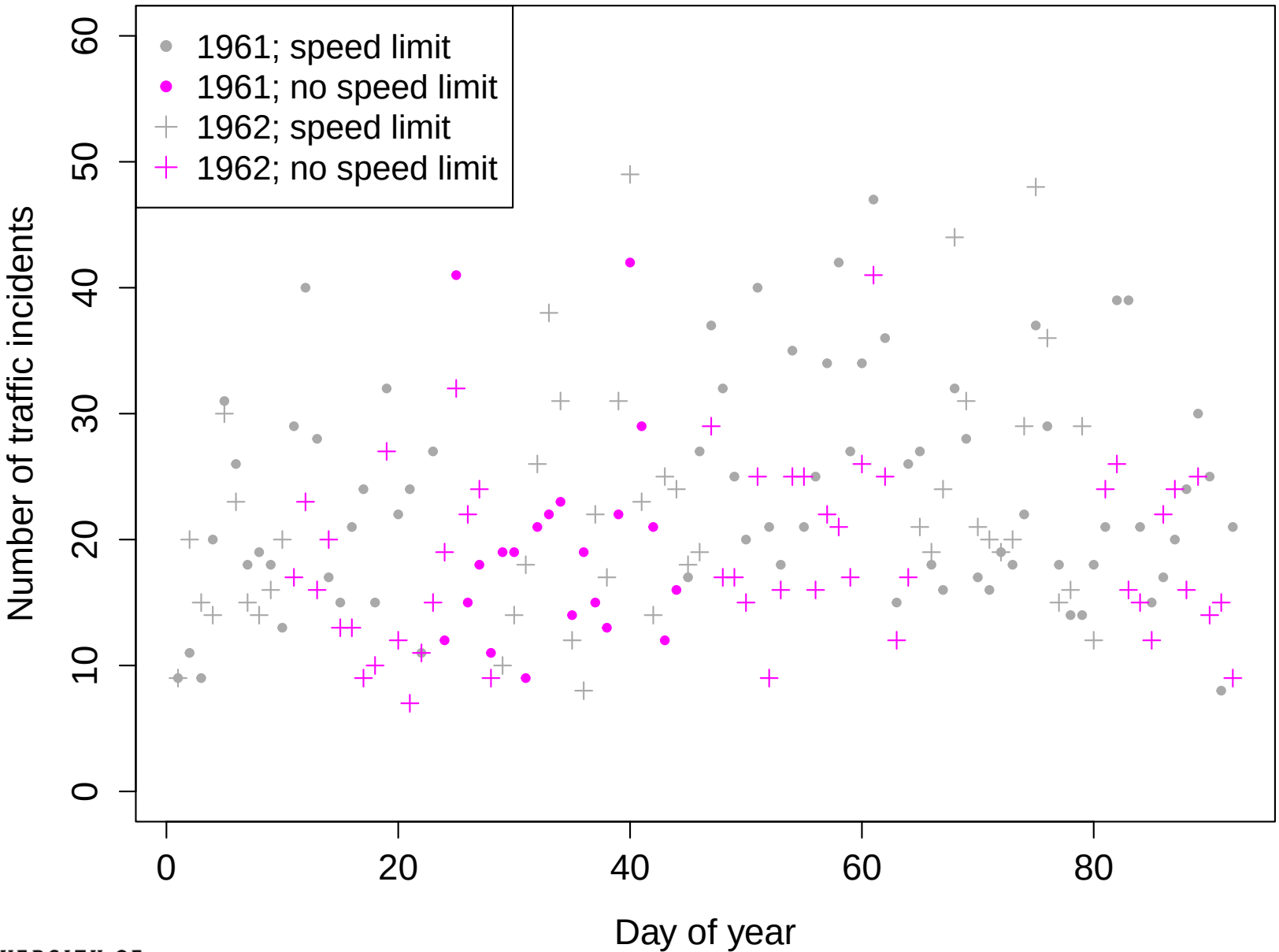
```
binom_chicken <- "
data {
  int<lower=1> N;                  // number of months
  int<lower=0> y[N];               // successes
  int<lower=0> n[N];               // trials
  int<lower=1,upper=N> month[N];
}

parameters {
  real alpha;                      // baseline (month 1)
  vector[N-1] delta;             // differences vs baseline
}

transformed parameters {
  vector[N] beta;
  beta[1] = 0;                     // reference month
  beta[2:N] = delta;
}

model {
  alpha ~ normal(0, 2); delta ~ normal(0, 2);  // priors

  // likelihood
  for (i in 1:N) {
    y[i] ~ binomial_logit(n[i], alpha + beta[month[i]]);
  }
}
"
```

Logit binomial vaccination rate effects relative to January



- Vaccination rates in March, July and Nov are higher than in January
- Probability distributions of pairwise comparisons obtained solidly ... ?

Swedish traffic incidents in 1961-62; (?) speed limit applied



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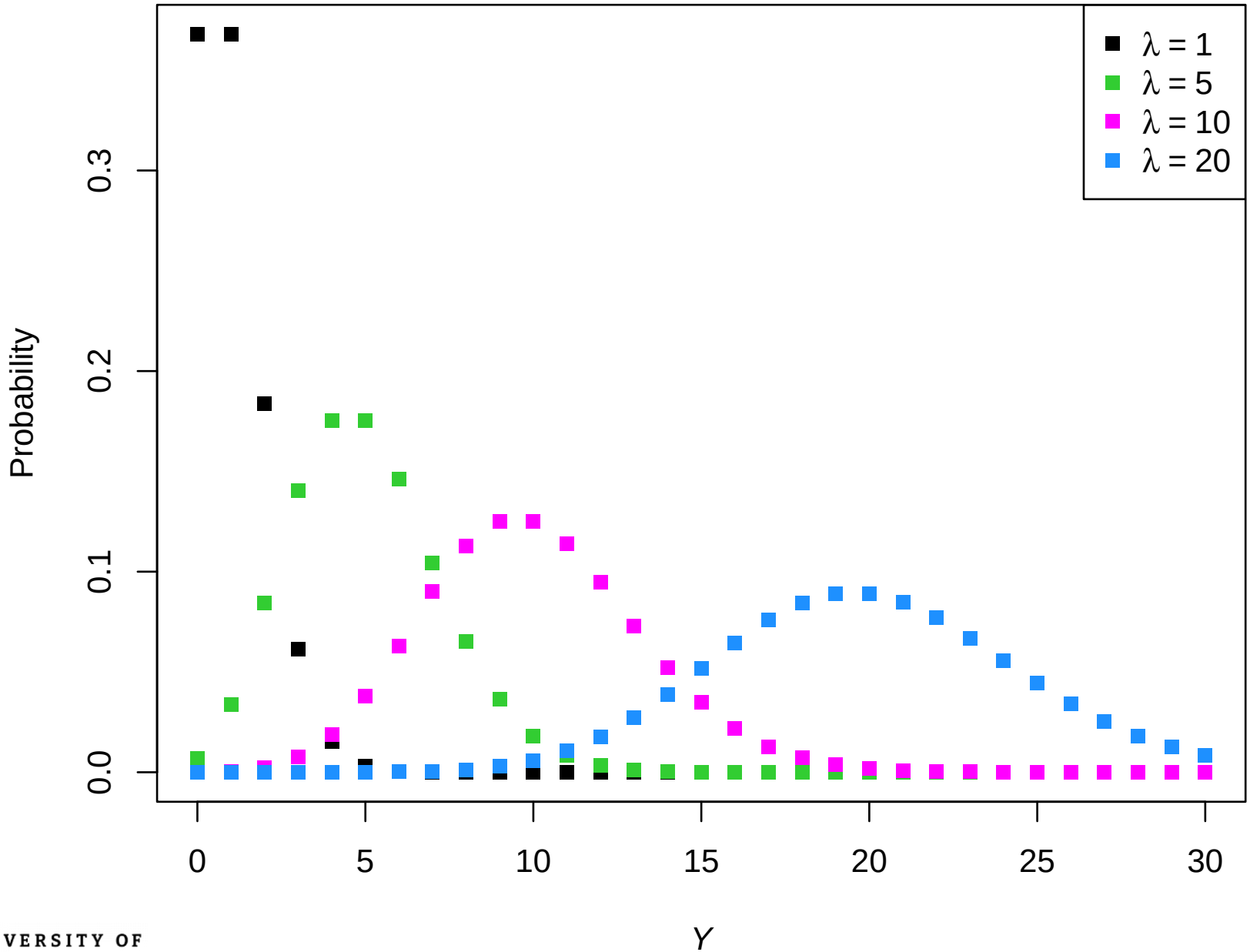
Poisson distribution with mean and variance parameter λ



Discrete/Count data modeling using poisson distributions

Poisson is a special case of the Negative-Binomial distribution

$$P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{3}$$

- For discrete values $k = \{0, 1, 2, ...\}$
- λ is a continuous parameter
- $y_i \sim \text{NegBinomial}(\mu_i, \phi);\ \text{var}(y_i) = \mu_i + \mu_i^2/\phi$

Discrete/Count data modeling using poisson distributions

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Fit traffic incident counts: $y \sim \text{Poisson}(\lambda = ...)$

Simple poisson model coding in RSTAN

```
library(rstan)
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)

pois_simple <- "
  data {
    int<lower=0> N;
    int<lower=0> y[N];
    real prior_lambda;
  }
  parameters {
    real<lower=0> lambda;
  }
  model {
    lambda ~ uniform(prior_lambda);
    y ~ poisson(lambda);
  }
"

# Initialize STAN model
pois_simple_mod <- stan_model(model_code=pois_simple, model_name="pois_simple")

# Assign data object and sample from Poisson model
data(Traffic, package = "MASS") # Load traffic data from MASS package
Traffic_dat01 <- list(N=nrow(Traffic), y=Traffic$y, prior_lambda=20) # Assign traffic data to a list
Traffic_dat02 <- list(N=nrow(Traffic), y=Traffic$y, prior_lambda=50)# Adjust prior
```

Poisson model output from RSTAN

```
pois_simple.rs1 <- sampling(pois_simple_mod, data=Traffic_dat01, iter=10e3, chains=4, thin=10)
pois_simple.rs2 <- sampling(pois_simple_mod, data=Traffic_dat02, iter=10e3, chains=4, thin=10)

write.csv(print(summary(pois_simple.rs1, pars=c("lambda","lp__"))$summary),
          "pois_simple_rs01.csv")
```

```
##             mean  se_mean      sd     2.5%      25%      50%      75%    97.5%   n_eff     Rhat
## lambda    19.930 0.002016 0.06625   19.754   19.897   19.952   19.979   19.998 1079.63 0.99982
```

```
write.csv(print(summary(pois_simple.rs2, pars=c("lambda","lp__"))$summary),
          "pois_simple_rs02.csv")
```

```
##             mean  se_mean      sd     2.5%      25%      50%      75%    97.5%   n_eff     Rhat
## lambda    21.540 0.009663 0.34206   20.879   21.304   21.528   21.760   22.239 1282.77 1.00035
```

Generalized linear prediction and the log-link function

Linear prediction

$$y \sim \text{Poisson}(\eta = \boldsymbol{X}\beta)$$

Generalized linear prediction and the log-link function

Linear prediction

$$y \sim \text{Poisson}(\eta = \boldsymbol{X}\beta)$$

Log-link function

- Instead of "raw linear prediction", use a function to avoid negative values:

$$y \sim \text{Poisson}(\exp(\boldsymbol{X}\beta))$$

$$\text{or: } \log(E[y]) = \boldsymbol{X}\beta$$

Log-link linear prediction model coding in STAN

```
pois_linpred <- "
data {
  int<lower=1> N; int<lower=1> K;
  int<lower=0> y[N];
  matrix[N,K] x;
  real<lower=0> prior_sd;
}

parameters {
  real alpha;
  vector[K] beta;
}

transformed parameters {
  vector[N] eta;
  eta = alpha + x * beta;  // linear predictor
}

model {
  // priors
  alpha ~ normal(3, prior_sd);
  beta ~ normal(0, prior_sd);

  y ~ poisson_log(eta);   // log link
}
"

pois_linpred_mod <- stan_model(model_code=pois_linpred, model_name="pois_linpred")
```

Generalized linear Poisson model and over-dispersion

Variance in the counts exceeds the expectated value of a Poisson

- Over-dispersion model with latent variable l and residual e_{res}:

$$\text{For: } \log(E[y]) = \boldsymbol{X}\beta$$

$$\text{use: } y \sim \text{Poisson}(\exp(l = \eta + e_{\text{res}})) \text{ with } e_{\text{res}} \sim N(0, \sigma_e)$$

Generalized linear Poisson model and over-dispersion

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- Over-dispersion model with latent variable l and residual e_{res}:

$$\text{For: } \log(E[y]) = \boldsymbol{X}\beta$$

$$\text{use: } y \sim \text{Poisson}(\exp(l = \eta + e_{\text{res}})) \text{ with } e_{\text{res}} \sim N(0, \sigma_e)$$

(Non)-centered parametrization: e_{res} and σ_e strongly (in)dependent

$$\frac{e_{\text{res}}}{\sigma_e} = e_{\text{raw}} \sim N(0,1),\ e_{\text{res}} = \sigma_e \cdot e_{\text{raw}} \tag{4}$$

- e_{raw} is independent of σ_e (prevents sampling issues)

Over-dispersed log-link linear prediction model coding

```
pois_linpred_disp <- "
data {
  int<lower=1> N; int<lower=1> K;
  int<lower=0> y[N]; matrix[N,K] x;
  real<lower=0> prior_sd;
}

parameters {
  real alpha; vector[K] beta;
  vector[N] e_raw; real<lower=0> sigma_e;
}

transformed parameters {
  // linear predictor and residual and latent variable
  vector[N] e_res; vector[N] eta; vector[N] latent;

  e_res = sigma_e * e_raw; eta = alpha + x * beta;
  latent = eta + e_res;
}

model {
  alpha ~ normal(3, prior_sd); beta ~ normal(0, prior_sd);  // normal priors
  e_raw ~ normal(0, 1); sigma_e ~ normal(0, 0.1);  // (half)-normal priors

  y ~ poisson_log(latent);   // log link function
}
"
pois_linpred_disp_mod <- stan_model(model_code=pois_linpred_disp, model_name="pois_linpred_disp")
```

Linear prediction using (over-dispersed) model outputs

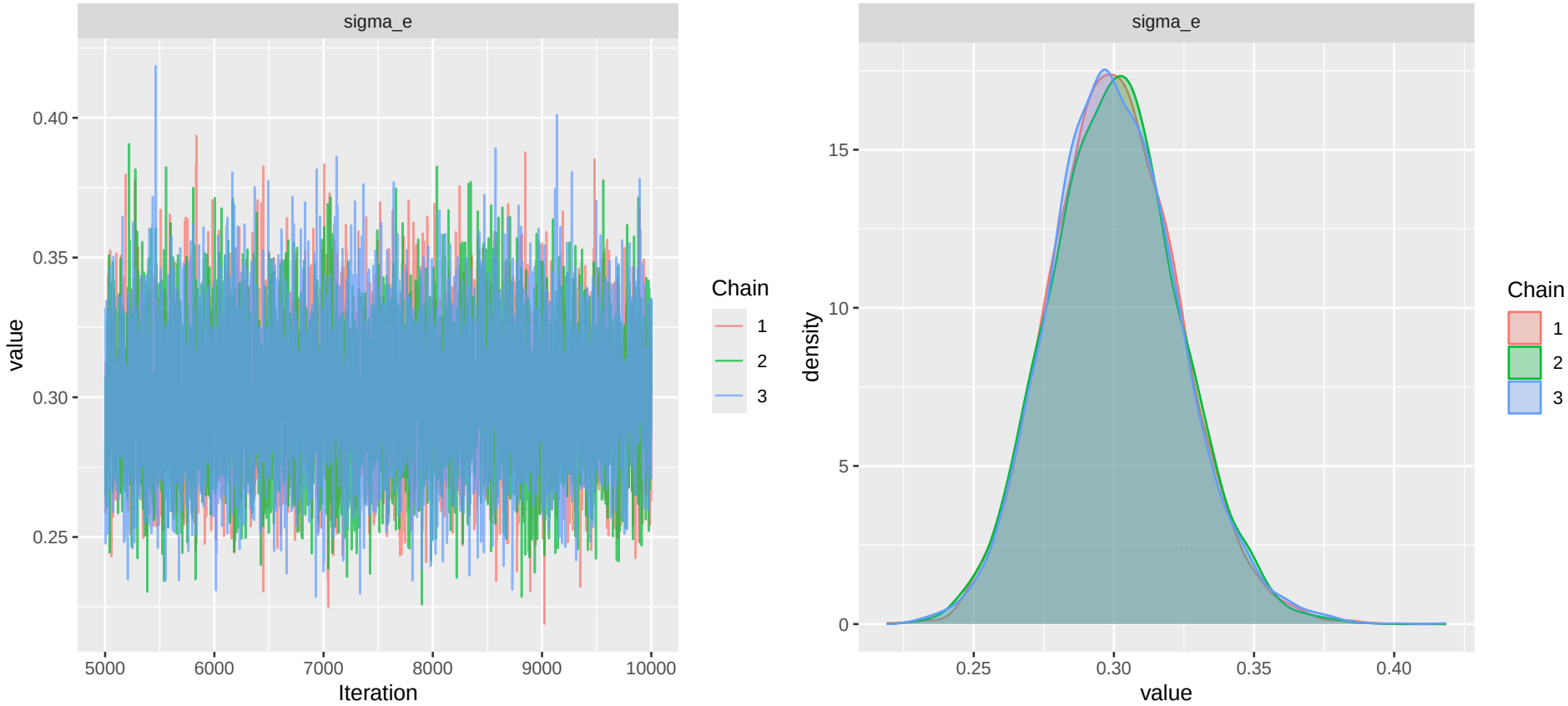
```
# Initialize data and assign new data list object
data(Traffic, package = "MASS")
Traffic_dat_list <- list(N=nrow(Traffic), K=3, y=Traffic[,4], x=Traffic[,-4])

Traffic_dat_list$x$limit <- as.integer(Traffic_dat_list$x$limit)-1
Traffic_dat_list$x$year <- as.integer(Traffic_dat_list$x$year)

# Multiple generalized linear regression without dispersion
Traffic_dat_list$prior_sd <- 1 # prior SD for alpha and beta
pois_linpred.rs1 <- sampling(pois_linpred_mod,  data=Traffic_dat_list, iter=10e3, chains=4, thin=10)
pois_linpred_d.rs1 <- sampling(pois_linpred_disp_mod, data=Traffic_dat_list, iter=10e3, chains=4, thin=10)

print(summary(pois_linpred.rs1, pars=c("alpha","beta","lp__","eta"))$summary)
```

```
#              mean  se_mean    sd      2.5%       25%       50%       75%      97.5%   n_eff    Rhat
# alpha      3.0987   0.0013 0.0575    2.9856    3.0595    3.0993    3.1369    3.2126 1890.763 1.0003
# beta[1]   -0.0570   0.0007 0.0331   -0.1213   -0.0793   -0.0574   -0.0345    0.0093 1953.237 1.0000
# beta[2]    0.0025   0.0000 0.0006    0.0013    0.0020    0.0025    0.0029    0.0037 3614.539 0.9998
# beta[3]   -0.1748   0.0008 0.0357   -0.2436   -0.1990   -0.1754   -0.1504   -0.1044 2234.596 0.9998
# eta[1]     3.0442   0.0007 0.0373    2.9720    3.0188    3.0437    3.0697    3.1185 2534.137 1.0003
```

```
print(summary(pois_linpred_d.rs1, pars=c("alpha","beta","sigma_e","lp__","eta"))$summary)
```

```
#              mean se_mean      sd      2.5%       25%       50%       75%      97.5%   n_eff    Rhat
# alpha      3.0587  0.0016  0.0997    2.8626    2.9922    3.0588    3.1264    3.2542 4046.491 1.0004
# beta[1]   -0.0653  0.0009  0.0575   -0.1794   -0.1038   -0.0658   -0.0272    0.0474 3761.128 1.0003
# beta[2]    0.0026  0.0000  0.0011    0.0005    0.0019    0.0025    0.0033    0.0046 5540.265 1.0007
# beta[3]   -0.1708  0.0009  0.0602   -0.2880   -0.2105   -0.1721   -0.1311   -0.0520 4885.349 1.0004
# sigma_e    0.3005  0.0003  0.0226    0.2584    0.2849    0.2995    0.3151    0.3483 5636.587 0.9997
# eta[1]     2.9960  0.0009  0.0637    2.8692    2.9533    2.9955    3.0395    3.1202 5060.194 1.0006
# eta[2]     2.9985  0.0009  0.0629    2.8734    2.9565    2.9981    3.0415    3.1211 5054.191 1.0006
```

Linear prediction (overdispersed) model outputs



- Over-dispersion: σ_e has shifted away from zero, whereas $\sigma_e = 0$ implies the response conforms to a standard Poisson

Generate posterior predictive distributions in RSTAN?

Finish the STAN code for the models with and without overdispersion parameter:

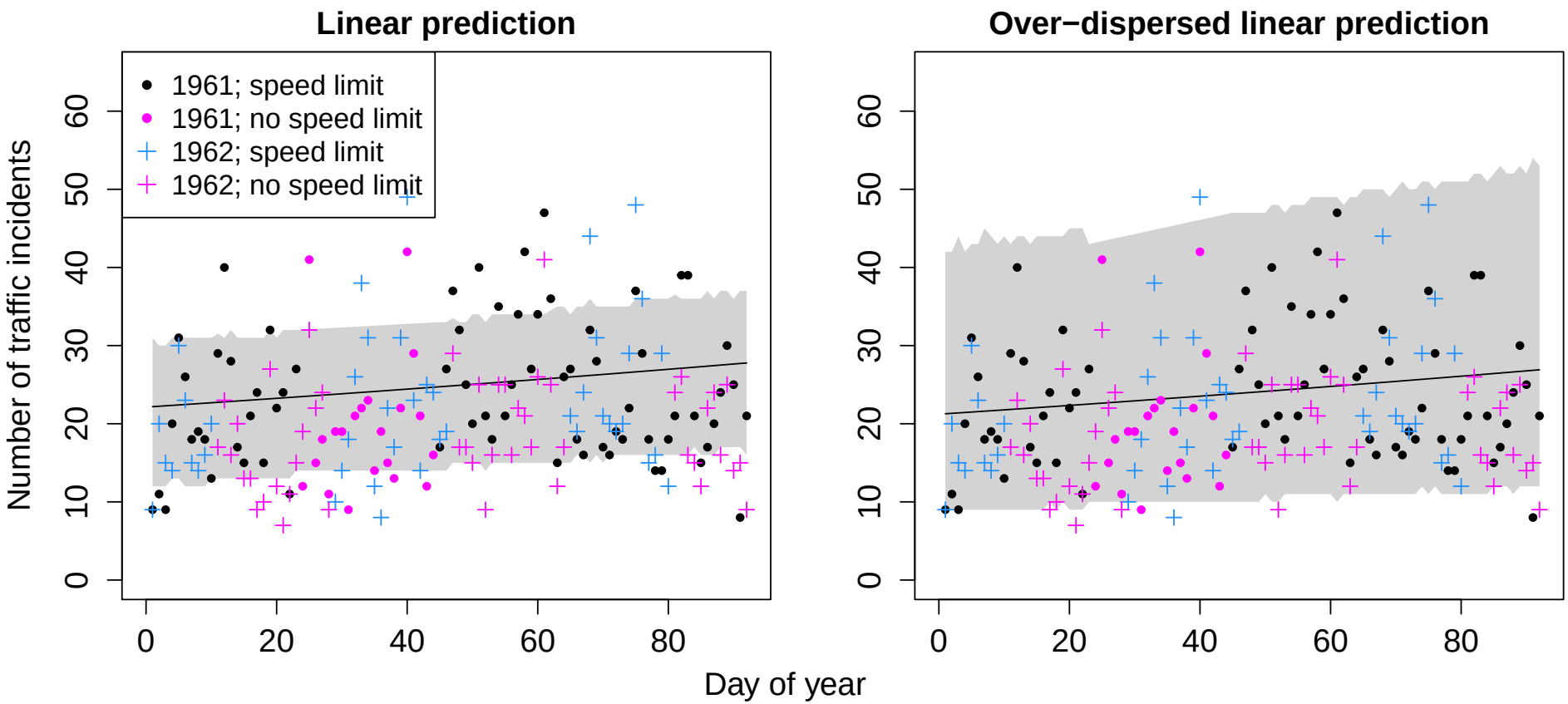
ypred[n] = poisson_log_rng(... ... ...);

Poisson posterior predictive distributions - STAN code

```
poisson_linpred <- "
  ...

generated quantities {
  int ypred[N]; vector[N] eta_rs;
  eta_rs = alpha + x[,2] * beta[2];

  for (n in 1:N){
    ypred[n] = poisson_log_rng(eta_rs[n]);
  } // End for loop n
}
"
```

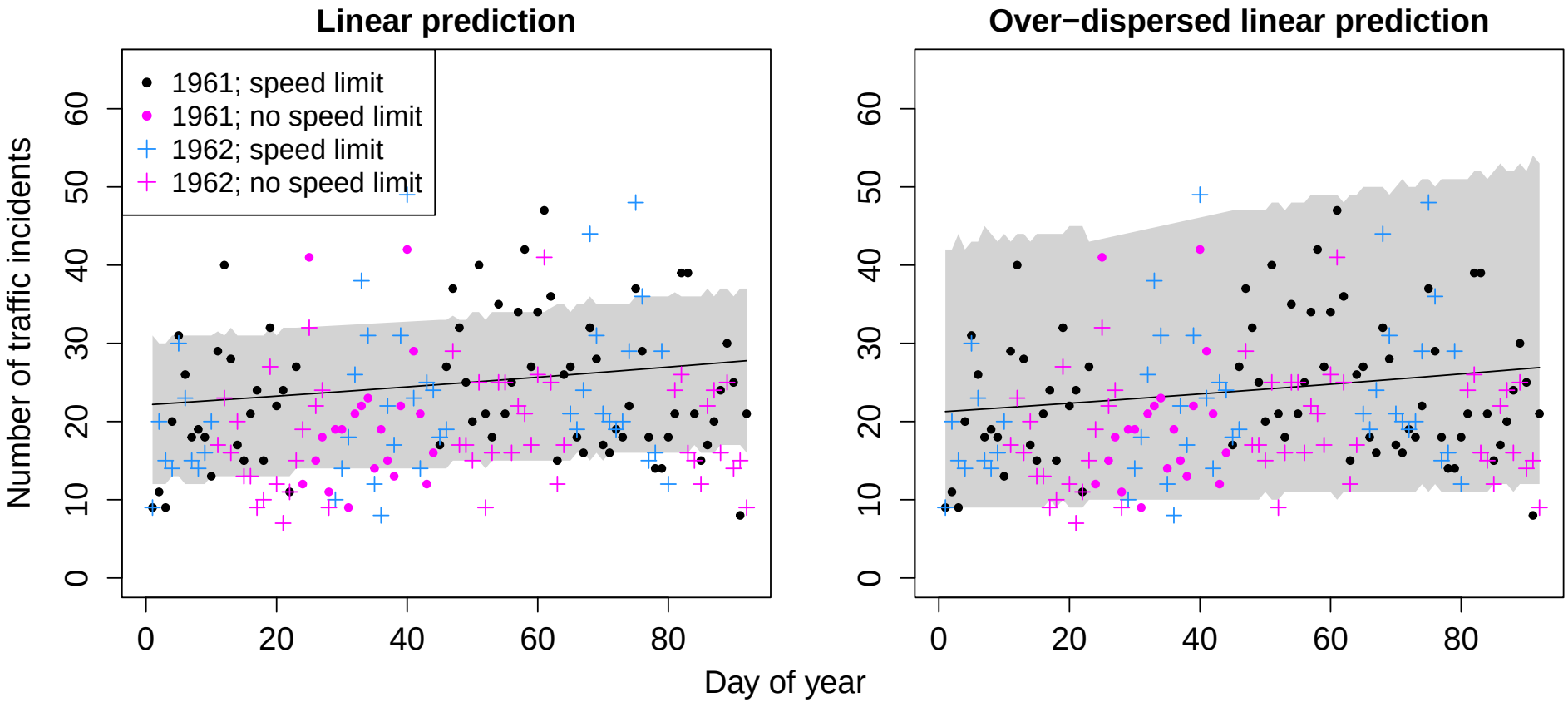
```
poisson_linpred_disp <- "
  ...

generated quantities {
  int ypred[N]; real pred_res; vector[N] eta_rs;
  eta_rs = alpha + x[,2] * beta[2];

  for (n in 1:N){
    pred_res = normal_rng(0, sigma_e);
    ypred[n] = poisson_log_rng(fmin(eta_rs[n],20) + pred_res);
  } // End for loop n
}
"
```

Posterior predictive distributions of Poisson models



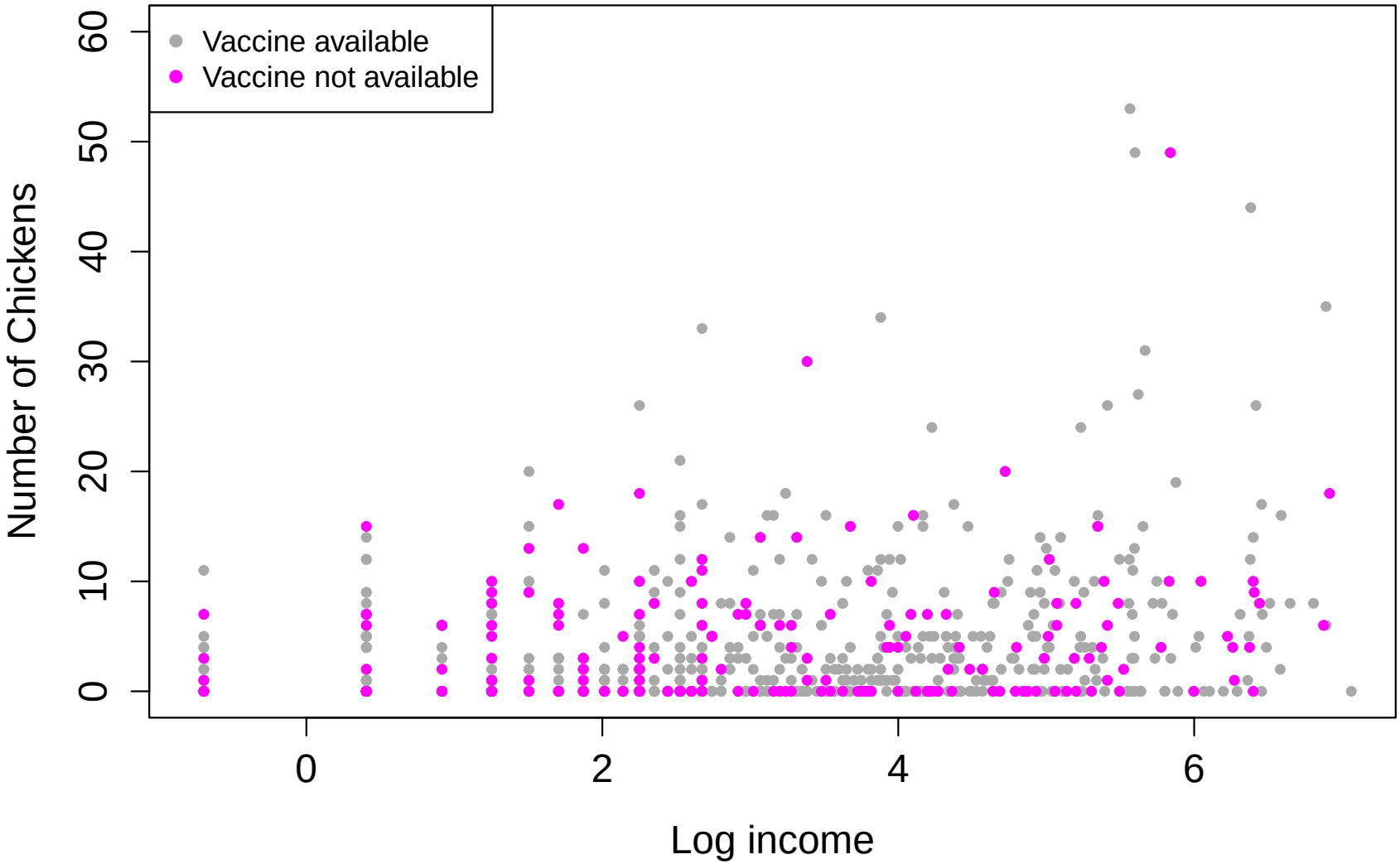
- Over-dispersion model obviously fits the data the best

Posterior predictive distributions of Poisson models



- Over-dispersion model obviously fits the data the best
- Elaborate traffic incident modeling in *MCMCglmm* course notes

Chickens per household after vaccination campaigning



- Poisson modeling: Do vaccine availability and income predict the number of chickens at the end of the vaccination campaign?

De Bruyn et al., (2017)

Poisson modeling of chicken count data

Generalized linear prediction

$$y_i \sim \text{Poisson}(\exp(\eta_i + \epsilon_i)); \qquad \epsilon_i \sim N(0, \sigma_e) \tag{5}$$

$$\eta_i = \alpha_{\nu,i} + \beta x_i \quad \text{for vaccination effect } \alpha_\nu \text{ and income predictor } \beta \tag{6}$$

Poisson modeling of chicken count data

Generalized linear prediction

$$y_i \sim \text{Poisson}(\exp(\eta_i + \epsilon_i)); \qquad \epsilon_i \sim N(0, \sigma_e) \tag{5}$$

$$\eta_i = \alpha_{\nu,i} + \beta x_i \quad \text{for vaccination effect } \alpha_\nu \text{ and income predictor } \beta \tag{6}$$

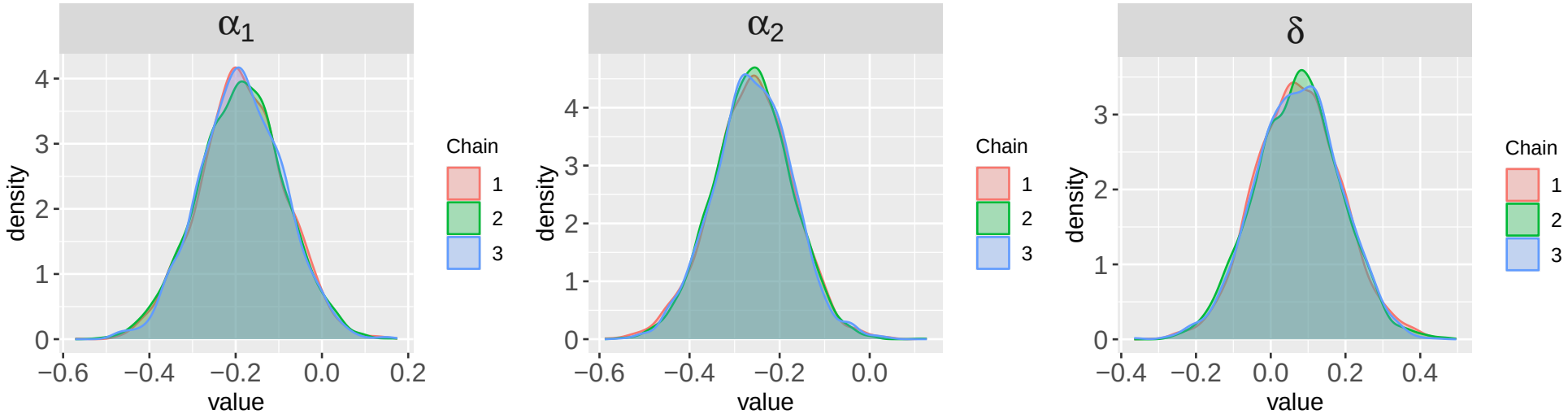
Generalized linear hierarchical prediction - include village effects

$$y_{ij} \sim \text{Poisson}(\exp(\eta_{ij} + u_j + \epsilon_{ij})); \qquad u_j \sim N(0, \sigma_u) \tag{7}$$

Poisson log-linear prediction of chicken counts in RSTAN

```
poisson_linpred_disp <- "
data {
  int<lower=1> N; int<lower=0> y[N]; vector[N] x; // data points N
  int<lower=1> M; int<lower=1,upper=M> v[N];      // vacc available as 1's and 2's in v
  real<lower=0> prior_sd; real prior_alpha; real prior_beta; // prior parameters
}

parameters {
  vector[M] alpha; real beta;
  vector[N] e_raw;  real<lower=0> sigma_e;
}

transformed parameters {
  // linear predictor and residual and latent variable
  vector[N] e_res; vector[N] eta; vector[N] latent;

  e_res = sigma_e * e_raw; eta = alpha[v] + x * beta;
  latent = eta + e_res;
}

model {
  alpha ~ normal(prior_alpha, prior_sd);
  beta ~ normal(prior_beta, prior_sd);
  e_raw ~ normal(0, 1); sigma_e ~ normal(0, 2);  // half-normal

  y ~ poisson_log(latent);  // log link
}
"
```

Poisson log-linear vaccination effect on chicken counts

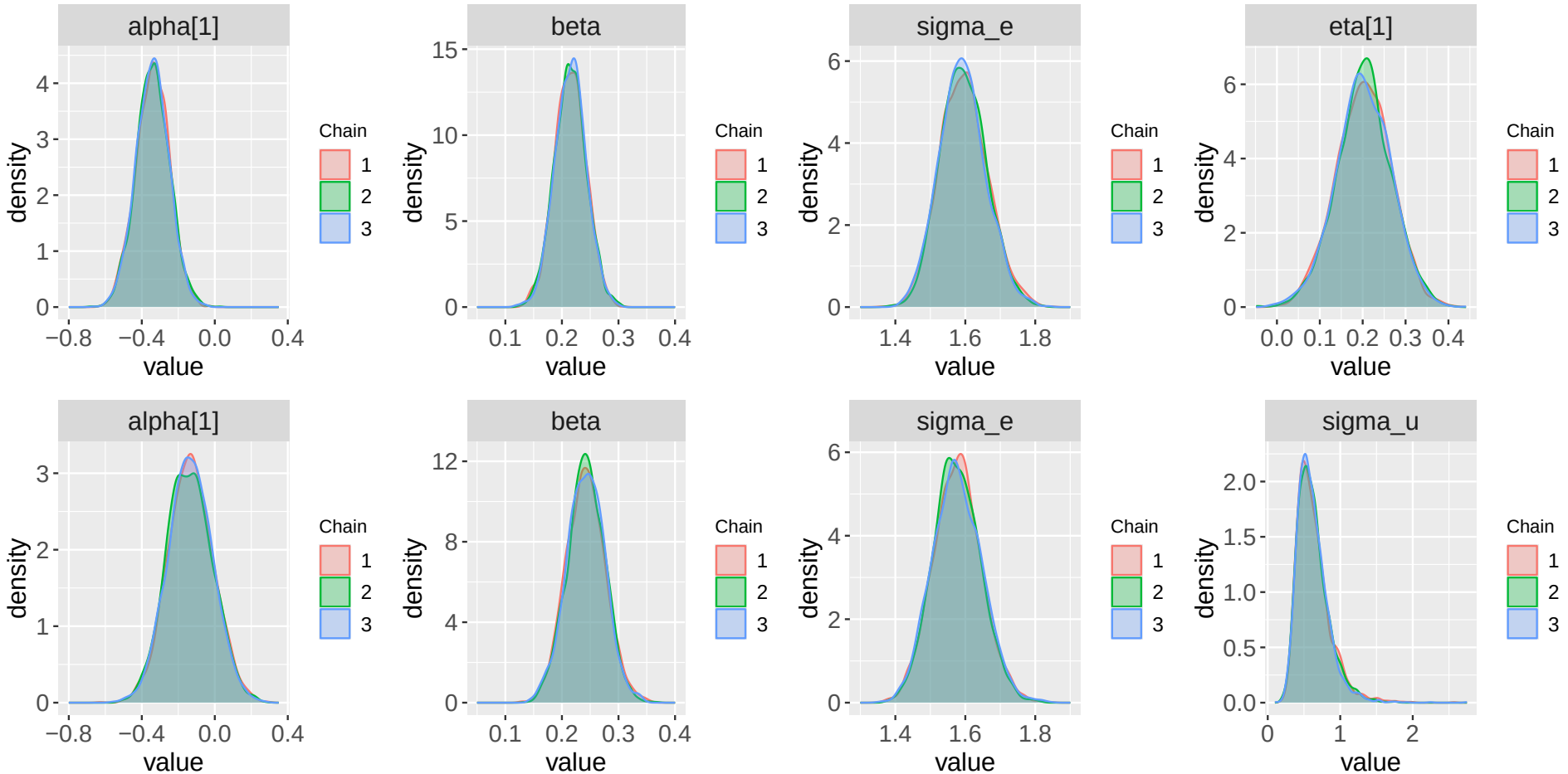


- Credible interval of δ, which is $\alpha_1 - \alpha_2$, contains zero
- No vaccination effect on number of chickens

Poisson hierarchical prediction of chicken counts in RSTAN

```
poisson_linpred_disp_hier <- "
data {
  int<lower=1> N; int<lower=0> y[N]; vector[N] x; // data points N
  int<lower=1> M; int<lower=1,upper=M> v[N]; // vacc available as 1's and 2's in v
  int<lower=1> K; int<lower=1,upper=K> vill[N]; // villages K
  real<lower=0> prior_sd; real prior_alpha; real prior_beta;
}

parameters {
  vector[M] alpha; real beta;
  vector[N] e_raw; vector[K] u;
  real<lower=0> sigma_u; real<lower=0> sigma_e;
}

transformed parameters {
  // linear predictor and residual and latent variable
  vector[N] eta; vector[N] e_res; vector[N] latent;

  e_res = sigma_e * e_raw; eta = alpha[v] + x * beta + u[vill];
  latent = eta + e_res;
}

model {
  alpha ~ normal(prior_alpha, prior_sd); beta ~ normal(prior_beta, prior_sd);
  e_raw ~ normal(0, 1);  u ~ normal(0, sigma_u);
  sigma_u ~ cauchy(0, 0.5); sigma_e ~ cauchy(0, 0.5);  // half-cauchy

  y ~ poisson_log(latent);  // log-link
} "
```

Poisson (hierarchical) model for over-dispersed counts



Generate posterior predictive distributions in RSTAN?

Finish the STAN code:

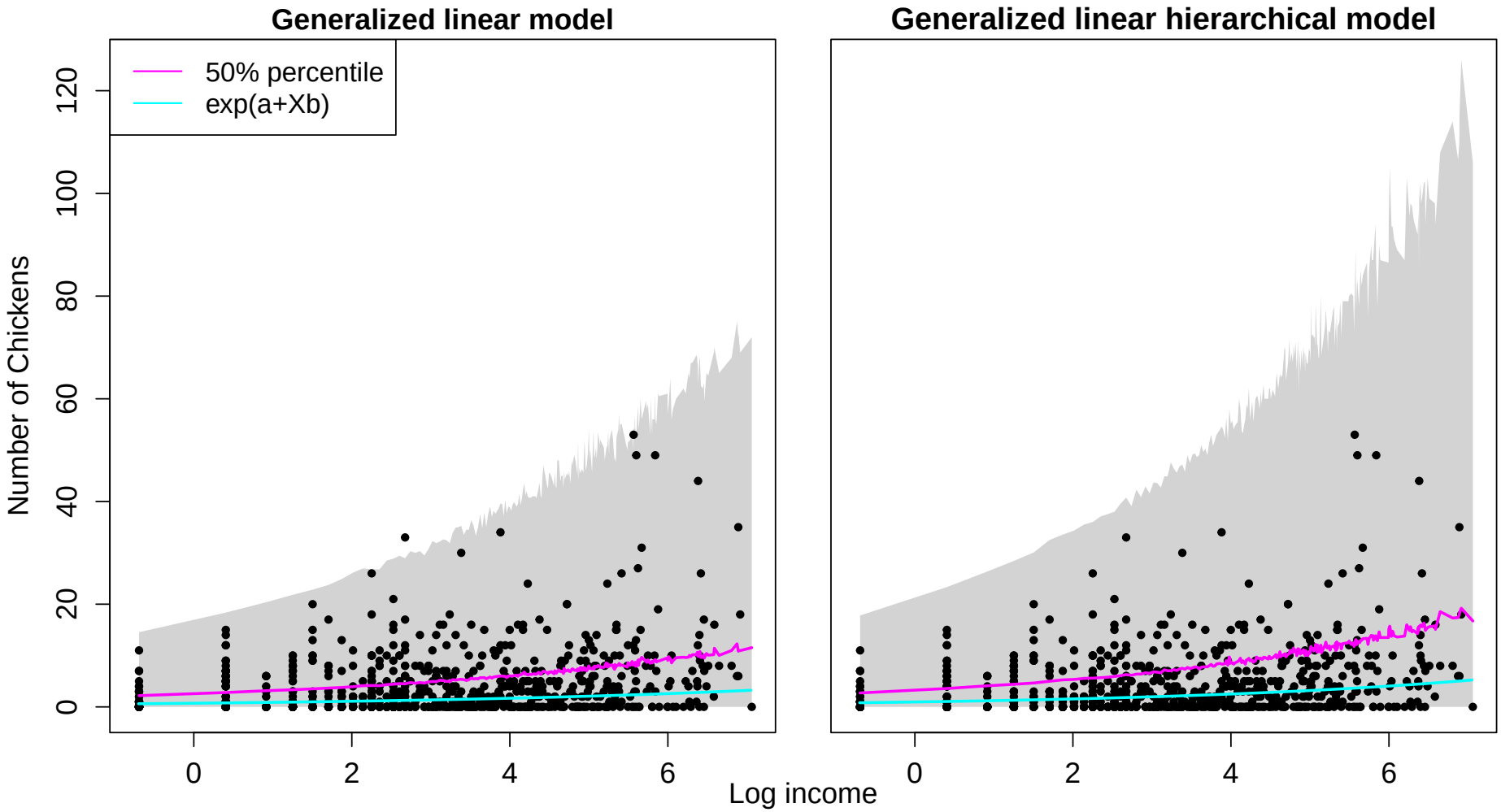
ypred[n] = poisson_log_rng(... ... ...);

Generate posterior predictive distributions in RSTAN!

```
 . . .

generated quantities {
  int ypred[N];
  vector[N] eta_rs;

  eta_rs = alpha[v] + x * beta;

  for (n in 1:N){
    real pred_res = normal_rng(0, sigma_e);
    ypred[n] = poisson_log_rng(fmin(eta_rs[n] + pred_res, 20)); // log(lambda) <= 20
  } // End for loop n
}"
```

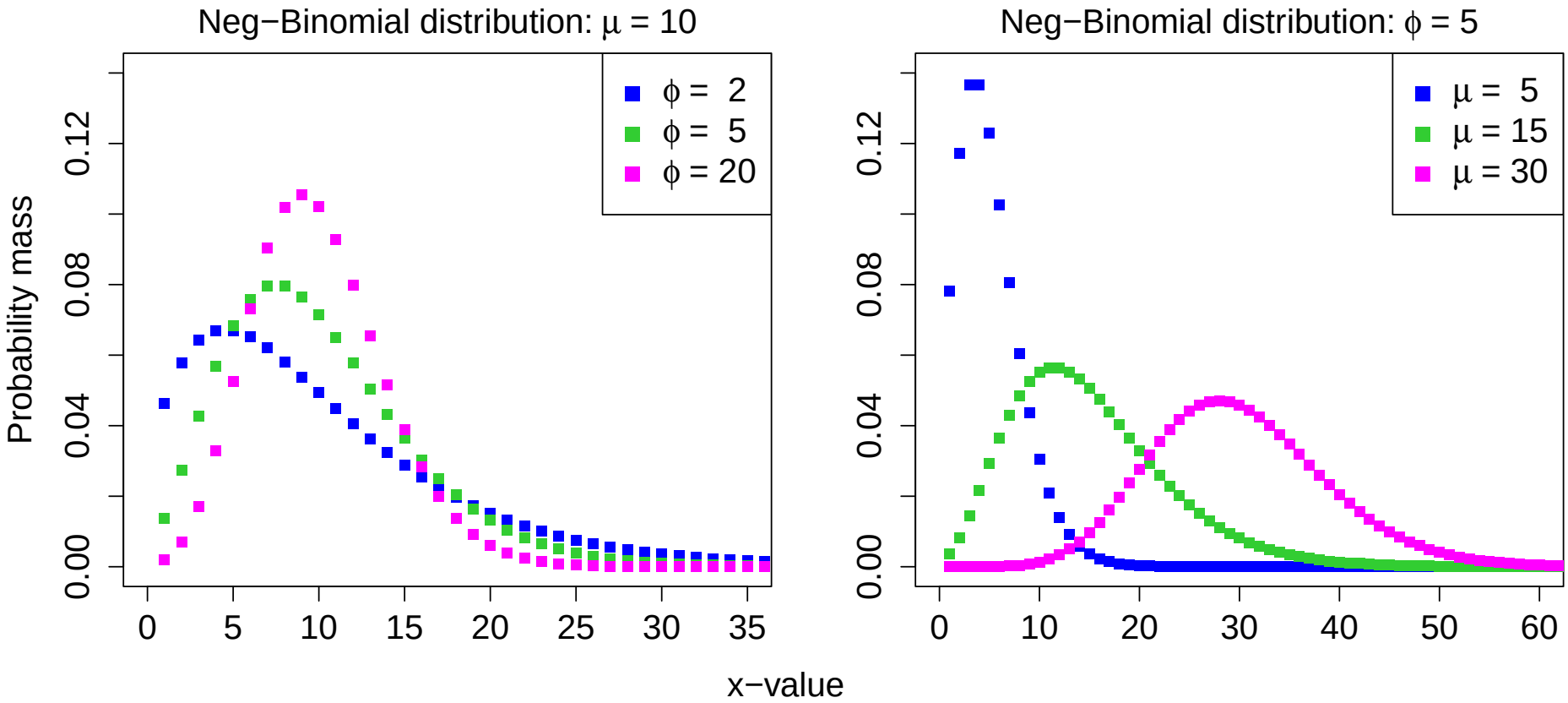
```
 . . .

generated quantities {
  int ypred[N];
  vector[N] eta_rs;

  eta_rs = alpha[v] + x * beta;

  for (n in 1:N){
    real pred_u = normal_rng(0, sigma_u);
    real pred_res = normal_rng(0, sigma_e);
    ypred[n] = poisson_log_rng(fmin(eta_rs[n] + pred_u + pred_res, 20)); // log(lambda) <= 20
  } // End for loop n
}"
```

Log-link Poisson posterior predictive distribution



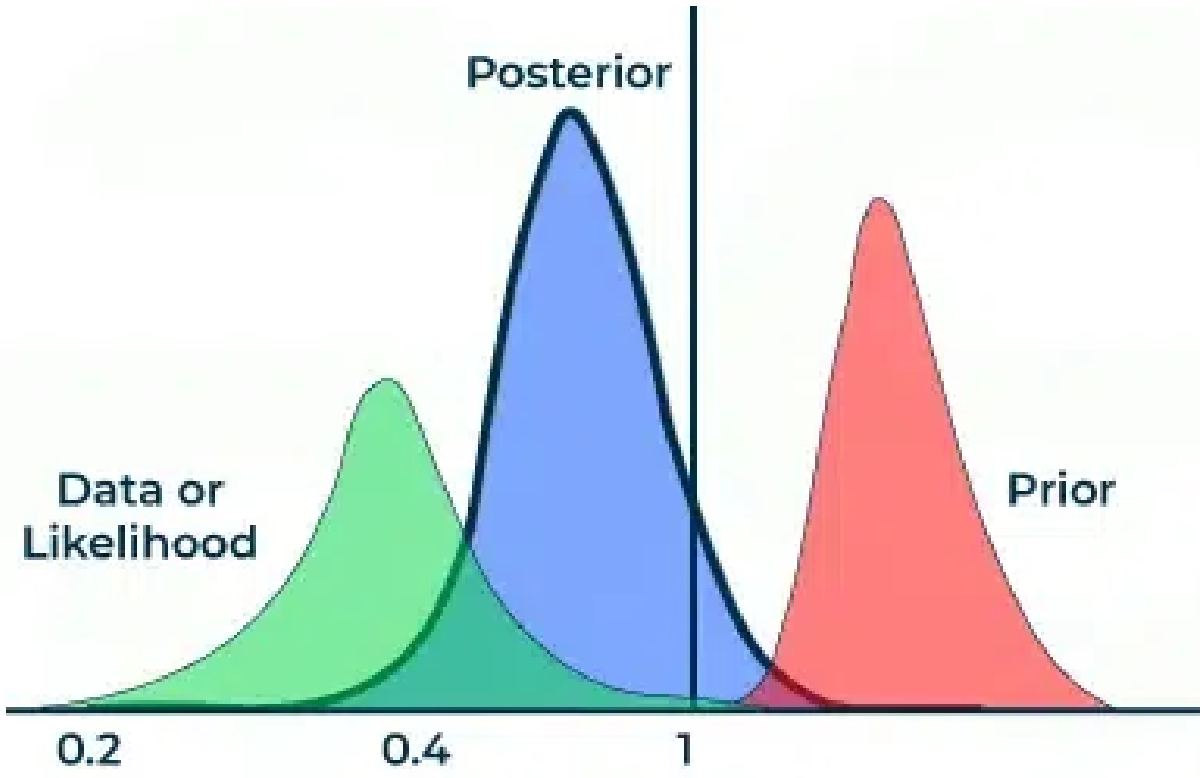
- Hierarchical model has larger noise/over-dispersion non-linearity
- Hierarchical model is over-parametrized

General case: Negative Binomial distribution



- Over-dispersion is assumed by definition
- Commonly used distribution in genomics and transcriptomics read count analysis

Thank you for your attention!



Questions?

Exercise lecture 5

- Consider the litter size data sets: a) parity effects in pigs, b) breed effects in pigs, c) breeds effects in dogs
- Fit the various poisson models to these datasets
- Consider various priors and generate posterior predictive distributions
- Evaluate breed and parity effects and over-dispersion
- Recode the overdispersed-poisson model as a negative-binomial, fit to one of the 3 datasets and the Traffic incident data, and evaluate the fits (use the posterior (predictive) distributions, etc)